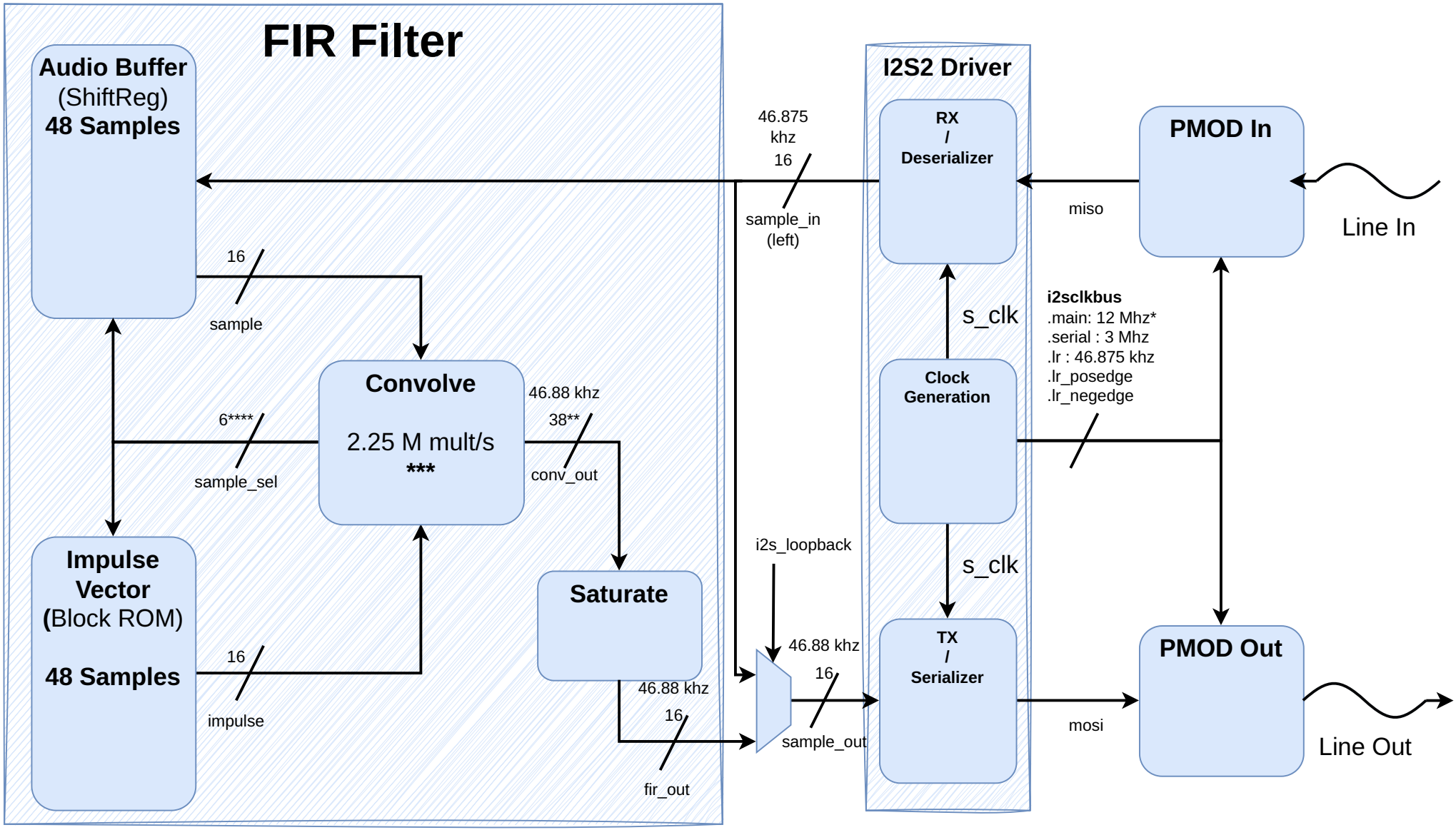
Top Level Block Diagram



Notes:
*: Clocks are determined by m_clk: lr_clk=m_clk/256, s_clk=lr_clk*64
**: conv_out is 38 bits because you are adding fourty-eight 32-bit numbers
***: 2.25 million multiplies per second required, as such multiplier cannot exceed 12/2.25=5 clock cycles per operation.

Perhaps a more useful way to view this is as follows: we need 1 convolution every lr_clk cycle, thus, we have 256 m_clk cycles to finish the operation.
****: clog2(48 samples) = 6 bits